

GOVERNMENT OF INDIA — MARITIME ENVIRONMENTAL PROTECTION  
DIVISION  
HEADQUARTERS INDIAN COAST GUARD • NATIONAL TECHNICAL  
RESEARCH ORGANISATION (NTRO)

TECHNICAL EVIDENCE DOSSIER & STATUTORY NOTICE OF VIOLATION  
ISSUED UNDER SECTION 356E, MERCHANT SHIPPING ACT 1958 & MARPOL 73/78 ANNEX I

|                  |                                         |                        |                              |
|------------------|-----------------------------------------|------------------------|------------------------------|
| Dossier Ref:     | ICG-MEP-DOSSIER-20260905-GULF           | Date of Issue:         | 05 Sep 2026, 16:06 UTC       |
| Maritime Sector: | Gujarat Offshore / Northern Arabian Sea | Legal Classification:  | ILLEGAL DISCHARGE (CRIMINAL) |
| Target Vessel:   | MT NEPTUNE GLORY                        | IMO / MMSI:            | 9384721 / 636019482          |
| Flag State:      | Liberia                                 | Attribution Certainty: | 82.0% (CONFIRMED)            |

1. SATELLITE RADAR REMOTE SENSING (SAR) OBSERVATION

|                           |                                                                                          |                            |                                       |
|---------------------------|------------------------------------------------------------------------------------------|----------------------------|---------------------------------------|
| Sensor Platform:          | Sentinel-1A C-Band SAR                                                                   | Acquisition Timestamp:     | 2026-09-04 13:10:00 UTC               |
| Swath Mode & Pol:         | Interferometric Wide (IW) Swath / VV (Co-polarization optimal for sea surface roughness) | Ground Resolution:         | 10.0 meters/pixel                     |
| Observed Slick Centroid:  | 22.4296° N, 69.1667° E                                                                   | Total Spill Surface Area:  | 27.32 km²                             |
| Estimated Discharge Mass: | 601.0 Metric Tonnes                                                                      | Lookalike Rejection Score: | 65.0% Mineral Crude Confidence        |
| ADIOS Weathering State:   | Heavy Chocolate Mousse (Highly Viscous Emulsion)                                         | Evaporative Loss / Mousse: | 22.3% Evaporated / 75.0% Water Uptake |

2. REVERSE LAGRANGIAN HYDRODYNAMIC DRIFT ANALYSIS (ORIGIN ATTRIBUTION)

Using an advection-diffusion Lagrangian numerical formulation integrated with real-time oceanographic current vector fields ( $\Delta u=0.28$  m/s,  $\Delta v=0.18$  m/s) and Ekman surface windage (3.2% Stokes drift factor with 15° Coriolis deflection), the slick plume trajectory was hindcasted backwards in time. The particle convergence analysis resolved the **exact discharge origin point**:

|                                    |                                          |                           |                              |
|------------------------------------|------------------------------------------|---------------------------|------------------------------|
| Spill Origin Point ( $x_0, y_0$ ): | 22.35221° N, 69.09184° E                 | Calculated Slick Age:     | 10.5 Hours prior to SAR pass |
| Estimated Discharge Time:          | T – 10.5h (05 Sep 2026 approx 02:40 IST) | Total Hydrodynamic Drift: | 17.84 km displacement        |

3. AIS CORRELATION & KINEMATIC BEHAVIORAL ANOMALIES

Analysis of **6 vessels** operating in the sector filtered down to **6 corridor candidates**. Vessel **MT NEPTUNE GLORY** exhibited definitive spatio-temporal intersection with the discharge epicenter:

| Evaluation Metric               | Observed Telemetry Value            | Anomaly Assessment                                                       | Score Weight |
|---------------------------------|-------------------------------------|--------------------------------------------------------------------------|--------------|
| Closest Point of Approach (CPA) | 0.80 NM (1.48 km)                   | Exact spatial co-location with slick root                                | 90.6/100     |
| Temporal Coincidence            | $\Delta t = 1.00$ hours from origin | Precise temporal synchronization at $t_0$                                | 70.7/100     |
| Speed Over Ground (SOG)         | Drop: 14.7 → 5.1 kts                | Severe speed drop (14.7 -> 5.1 kts) matching illegal bilge purge profile | 100.0/100    |
| Vessel Risk Category            | Crude Oil Tanker (DWT: 158,000 T)   | High capacity oily water / slop separator                                | 95.0/100     |

4. NON-COOPERATIVE / DARK VESSEL RADAR SURVEILLANCE AUDIT

**TACTICAL RADAR TARGET INTERCEPTION:** CFAR point-target detection extracted **5 radar contacts** with metallic corner-reflector signatures ( $>+5$  dB  $\sigma^0$ ) that broadcast **ZERO AIS transponder telemetry**.

|                 |                     |                               |                       |
|-----------------|---------------------|-------------------------------|-----------------------|
| Radar Target ID | Position (Lat, Lon) | RCS ( $\sigma^0$ dB) / Length | Threat Classification |
|-----------------|---------------------|-------------------------------|-----------------------|

|                |                        |                   |                                             |
|----------------|------------------------|-------------------|---------------------------------------------|
| RADAR-TGT-2152 | 22.3927° N, 69.2884° E | 27.8 dB / ~141.4m | HIGH PRIORITY SUSPECT<br>(COVERT DISCHARGE) |
| RADAR-TGT-2153 | 22.3927° N, 69.2811° E | 26.5 dB / ~106.1m | HIGH PRIORITY SUSPECT<br>(COVERT DISCHARGE) |
| RADAR-TGT-8237 | 22.5193° N, 69.2383° E | 29.5 dB / ~201.2m | UNREGISTERED MARITIME<br>CONTACT            |

5. COASTAL HAZARD & BEACHING INTERCEPTION ALERT

CRITICAL SHORELINE ALERT: Hydrodynamic forecasting indicates slick impact along the coastline within 8.0 hours. Interception coordinates: 22.5648° N, 69.3181° E. Immediate mobilization of ICG Pollution Response (PR) Vessels and containment booms is ordered.

5. STATUTORY DIRECTIVES & DETENTION ORDER

NOTICE TO MASTER, OWNER, AND FLAG ADMINISTRATION:  
Pursuant to powers vested under Section 356J and Section 356E of the Merchant Shipping Act, 1958:  
1. **PORT STATE CONTROL INTERCEPTION:** The Principal Officer, Mercantile Marine Department (MMD), is directed to **DETAIN** vessel **MT NEPTUNE GLORY (IMO: 9384721)** upon entry into any Indian port or anchorage until mandatory oil record book (ORB Part I & II) inspection and bilge manifold swab sampling are concluded.  
2. **FINANCIAL SECURITY:** A maritime environmental lien of **INR 25,00,00,000 (Twenty-Five Crore Rupees)** is hereby levied to cover offshore containment, dispersant spraying, and coastal remediation costs.  
3. **FLAG STATE NOTIFICATION:** A copy of this satellite radar and AIS kinematic forensic package is dispatched to the International Maritime Organization (IMO) and the Maritime Administration of Liberia.

|                                                                                                             |                                                                                                             |
|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Forensic Chain-of-Custody SHA-256 Hash:<br>dbbaa41bc9333be454c7b51862e419efb0a8bfc0d278d71363a80e21790c26f0 | Authorized Signatory:<br>Inspector General (Operations & MEP)<br>Indian Coast Guard Headquarters, New Delhi |
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